

Zero Level

1. A coin is tossed once. Find the probability of getting a head.
2. In a single throw of two dice, find the probability of obtaining 'a total of 8'.
3. A coin is tossed once. Find the probability of getting a tail .
4. If S is the sample space of a certain experiment, what can you say about $P(S)$?
5. A coin is tossed twice. What is the probability that at least one tail occurs ?
6. A die is thrown. Find the probability of :
(i) getting a prime number
(ii) getting a number ≥ 3
(iii) getting a number ≤ 4
(iv) getting a number more than 6
(v) getting a number less than 6
(vi) getting a number 5
(vii) getting an odd number
(viii) getting a number between 3 and 6
(ix) getting a multiple of 3
7. If $2/11$ is the probability then an event will happen, what is the probability that it will not happen ?
8. In a simultaneous toss of two coins, find the probability of :
(i) exactly 2 tails (ii) exactly one tail (iii) no tail.
9. In a single throw of two dice, determine the probability of not getting the same number on the two dice.
10. In a single throw of two dice, find the probability of :
(i) getting a sum less than 6
(ii) getting a doublet of odd numbers
(iii) getting the sum as a prime number
11. In a single throw of two dice, find :
(i) P (an odd number on the first die and a 6 on the second)
(ii) P (a number greater than 3 on each side)
(iii) P (a total of 10)
(iv) P (a total greater than 8)
(v) P (a total of 9 or 11)
12. Three coins are tossed once. Find the probability of getting :
(i) 3 heads (ii) exactly 2 heads
(iii) at least 2 heads (iv) at most 2 heads
(v) no heads (vi) 3 tails
(vii) exactly 2 tails (viii) no tails
(ix) exactly one tail (x) a head on the first coin.
13. A card is drawn at random from a well-shuffled deck of 52 cards. Find the probability of getting :
(i) a king (ii) heart
(iii) a seven of heart (iv) a diamond
(v) a red card (vi) a face card
(vii) a club (viii) a non spade

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- (ix) a jack (x) a queen.
14. A card is drawn at random from a well-shuffled deck of 52 cards. If A is the event of getting a red card and the event B that a card is bearing a number greater than 2 but less than 9, find $P(A)$ and $P(B)$.
15. A card is drawn at random from a well-shuffled pack of 52 cards. What is the probability that the card bears a number greater than 3 and less than 10 ?
16. A bag contain 9 red and 12 white balls. One ball drawn at random. Find the probability that the ball drawn is red.
17. From a bag containing 7 red and 5 black marbles, one is drawn at random. What is probability of the marble being :
(i) black ? (ii) red ?
18. An urn contain 9 red, 7 white and 4 black balls. A ball is drawn at random. What is the probability that the ball drawn will be :
(i) red ? (ii) white ? (iii) black ? (iv) red or black
(v) white or black ? (vi) not red ? (vii) not black ?
19. A bag contain 4 white and 5 black balls. A ball is drawn at random from the bag. Find the probability that the ball drawn is white.
20. A bag contain 3 red balls bearing one of the numbers 1, 2 or 3 (one number on one ball); and 2 black balls bearing the numbers 4 or 6. A ball is drawn, its number is noted and the ball is replaced in the bag. Then another ball is drawn and its number is noted. Find the probability of drawing :
(i) 2 on the first draw and 6 on the second draw
(ii) a number ≤ 2 on the first draw and 4 on the second draw.
(iii) a total of 5.
21. If a letter is chosen at random from the English alphabet, find the probability that the letter is (i) a vowel
(ii) a consonant.
22. What is the probability that an ordinary year has 53 tuesday ?
23. What is the probability that a leap year has 53 sunday ?
24. 20 cards are numbered from 1 to 20. One card is then drawn at random. What is the probability that the number on the card will be :
(i) a multiple of 4 ? (ii) not a multiple of 4 ?
(iii) odd ? (iv) greater than 12 ?
(v) divisible by 5 ? (vi) not a multiple of 6 ?
25. In a lottery a person chooses six different numbers at random from 1 to 20, and if these six numbers match with the six numbers already fixed by the lottery committee, he wins the prize. What is the probability of winning the prize in the game ?
26. What are odds in favour of getting a spade if the card is drawn from a well-shuffled deck of 52 card ? What the odds in favour of getting a king ?
27. A card is drawn from a well shuffled deck of 52 cards. What are the odds in favour of getting a face card ?
28. The odds in favour of the occurrence of an event are

- 8 : 13; find the probability that the event will occur.
29. If the odds against the occurrence of an event are 4 : 7; find the probability of the occurrence of the event.
30. The odds in favour of occurrence of an event are 5 : 13. Find the probability that it will occur.
31. Two dice are thrown. Find :
(i) the odds in favour of getting the sum 6
(ii) the odds against getting the sum 7
32. If $\frac{5}{14}$ is the probability of occurrence of an event, find
(i) the odds in favour of its occurrence
(ii) the odds against its occurrence
33. If $\frac{7}{10}$ is the probability of occurrence of an event, what is the probability that it does not occur ?
34. What is the probability that in a group of two people, both will have the same birthday, assuming that there are 365 days in a year and no one has his/her birthday on 29th February ?
35. If $P(A) = \frac{3}{5}$ and $P(B) = \frac{1}{3}$, find :
 $P(A \text{ or } B)$, if A and B are mutually exclusive events.
36. If A and B are two events such that $P(A) = \frac{1}{4}$, $P(B) = \frac{1}{2}$ and $P(A \cap B) = \frac{1}{8}$, find :
(i) $P(A \text{ or } B)$ (ii) $P(\text{not } A \text{ and not } B)$.
37. If E_1 and E_2 are two events associated with a random experiment such that
 $P(E_2) = 0.35$, $P(E_1 \text{ or } E_2) = 0.85$ and
 $P(E_1 \text{ and } E_2) = 0.15$, find $P(E_1)$.
38. Let A and B be two events associated with a random experiment for which $P(A) = 0.4$, $P(B) = 0.5$ and $P(A \text{ or } B) = 0.6$. Find $P(A \text{ and } B)$.
39. If A and B are two mutually exclusive events such that $P(A) = \frac{1}{2}$ and $P(B) = \frac{1}{3}$, find $P(A \text{ or } B)$.
40. In a random experiment, let A and B be events such $P(A \text{ or } B) = 0.7$, $P(A \text{ and } B) = 0.3$ and $P(\bar{A}) = 0.4$, find $P(B)$ and $P(\bar{B})$.
41. If A and B are two events associated with a random experiment such that $P(A) = 0.25$, $P(B) = 0.4$ and $P(A \text{ or } B) = 0.5$, find the values of
(i) $P(A \text{ and } B)$ (ii) $P(A \text{ and } \bar{B})$
42. A and B are two events such that
 $P(A) = 0.42$, $P(B) = 0.48$ and $P(A \text{ and } B) = 0.16$.
Determine (i) $P(\text{not } A)$ (ii) $P(\text{not } B)$ (iii) $P(A \text{ or } B)$.
43. From a well-shuffled pack of 52 cards, a card is drawn at random. Find the probability of its being a king or a queen.
44. From a well-shuffled pack of 52 cards, a card is drawn at random. Find the probability of its being either a queen or heart.
45. A card is drawn at random from a well-shuffled deck 52 cards. Find the probability that the card is a :
(i) king or a red card (ii) club or a diamond
(iii) king or an ace (iv) spade or a club
(v) neither a heart nor a king.
46. Two dice are rolled once. Find the probability that :
(i) the number on the two dice are different
(ii) the total is at least 3
(iii) the total is 6
(iv) the total is 7 or 9
(v) neither a doublet nor a total of 9 appear.
47. A coin is tossed and a die is thrown. Find the probability that the outcome will be a head or a number greater than 4, or both.
48. A number is chosen from the number 1 to 100. Find the probability of its being divisible by 4 or 6.
49. A die is thrown twice. What is the probability that at least one of the two throws comes up with the number 4 ?
50. Two dice are thrown together. What is the probability that the sum of the numbers on the two faces is neither divisible by 3 nor by 4.
51. Two balls are drawn at random with replacement from a box containing 10 black and 8 red balls. Find the probability that :
(i) both balls are red
(ii) first ball is black and second is red.
(iii) one of them is black and other is red.
52. In a class, 30% of the students offered mathematics, 20% offered chemistry and 10% offered both. If a student is selected at random, find the probability that he has offered mathematics or chemistry.
53. The probability that a company executive will travel by plane is $\frac{2}{5}$ and that he will travel by train is $\frac{1}{3}$. Find the probability of his travelling by plane or train.
54. The probability that Bhuvish passes in English is $\frac{2}{3}$ and the probability that he passes in Hindi is $\frac{5}{9}$. If the probability of his passing both subject is $\frac{2}{5}$, find the probability that he will pass in atleast one of these subjects.
55. In a town of 6000 people, 1200 are over 50 years old and 2000 are females. It is known that 30% of the females are over 50 years. What is the probability that a randomly chosen individual from the town is either female or over 50 years ?
56. A coin is tossed once. Then, if it turns up a head a die is thrown once, if it turns up a tail it is tossed twice more. Describe
(i) the sample space S of the experiment.
(ii) the event A "that exactly one head occurs".
(iii) the event B "that at least two tails occur or a number greater than 4 occurs".
57. Three coins are tossed once. Let A be event "three heads show", B denote the event "two heads and one tail show", C denote the event "three tails show" and D denote the event "a head shows on the first coin".
Which events are :
(i) mutually exclusive ? (ii) simple ? (iii) compound ?